# Do LLMs Produce Regret-Like Language Under Deprivation Framing? A Prompt-Conditioned Behavioral Study Across Diverse Model Families

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***Abstract*—We investigate prompt-conditioned lexical activation in LLMs: under what prompting conditions does regret-like language systematically appear in generated text? Using a $3 \times 3$ controlled design with temperature as a continuous covariate ($N_{\text{total}} = 6{,}072$, 39 batches; 27 model variants spanning seven organizations), we treat regret-associated language as a behavioral signal of prompt-conditioned generation, not as evidence of subjective experience. The strongest finding is the ruminative persona effect ($z = 18.58$, $p < 0.001$ for embedding bias): system-prompt persona instructions are more reliable levers for affect-laden generation than user-prompt framing. A confirmatory linear mixed-effects model with scenario random intercepts shows that embedding-level semantic regret bias is significantly elevated under both deprivation ($\hat{\beta} = 0.162$, $z = 32.99$, $p < 0.001$) and counterfactual framing ($\hat{\beta} = 0.210$, $z = 40.60$, $p < 0.001$), revealing a *marker-type dissociation*—deprivation and counterfactual framings differ in lexical-layer signatures (CF rate both $p < 0.001$, suggesting counterfactual semantic priming does not require overt if-then linguistic structures. Lexical marker elevation under deprivation is confirmed for all markers: negemo rate ($p < 0.001$), regret-word rate ($p < 0.001$), and CF rate ($p < 0.001$). These patterns replicate directionally across all model variants tested. Code, stimuli, and outputs are publicly available.**

## KEYWORDS

LLM behavior, regret-like language, counterfactual framing, behavioral benchmarking, reproducibility

## I. INTRODUCTION

Humans express regret through counterfactual statements (*"If only I had..."*, *"Should have done..."*) and temporal re-evaluation of irreversible decisions [1], [2]. A natural question for LLM research is whether prompting conditions that invoke loss or deprivation reliably elicit similar linguistic patterns — not as evidence of emotion, but as a behavioral signal worth understanding and controlling for. We study this at the level of observable linguistic output, treating regret-like language as a prompt-conditioned behavioral signal rather than as evidence of subjective experience; accordingly, this is a *prompt-conditioned lexical activation study*, not a study of LLM internal emotional states.

This matters for two reasons. First, LLM outputs are increasingly used in emotionally sensitive applications; knowing which prompt structures activate regret-like language patterns has practical safety relevance [3]. Second, systematic behavioral mapping of LLM language responses to emotionally loaded prompts is methodologically important for alignment evaluation and anthropomorphism caution [4], [5].

Prior LLM behavioral studies have focused on theory-of-mind benchmarks [6], [7], but few have examined whether affect-laden framing reliably shifts the lexical distribution of generated text in a controlled, cross-model fashion. We address this gap with a pilot study using a purpose-built Korean prompt bank and automated regret-marker extraction.

**Research question:** Under what prompting conditions does regret-associated language systematically appear in LLM-generated text, and does this prompt-conditioned activation pattern replicate across proprietary and open-weight models?

## II. RELATED WORK

*a) Regret and counterfactual psychology.*

Regret is associated with upward counterfactual thinking about reversible or irreversible decisions [1], [8], [9]. Key operational definitions separate the linguistic manifestation of regret (counterfactual phrases, self-blame terms, temporal repair expressions) from its subjective experience [10]–[13], which is precisely the observable we target.

*b) LLM behavioral evaluation.*

Theory-of-mind and social cognition evaluations have demonstrated high sensitivity of LLM scores to prompt formulation and framing [5], [14], [15], motivating strict protocol control. We follow this tradition by using matched prompt conditions and fixed persona structures.

*c) Affective NLP and benchmarking.*

Emotion-annotated benchmarks [16], [17] and LLM emotion-evaluation studies [18], [19] show that label granularity and annotation design substantially affect measured performance. We adopt conservative claim framing: our lexical markers are exploratory instruments, not validated affect scales.

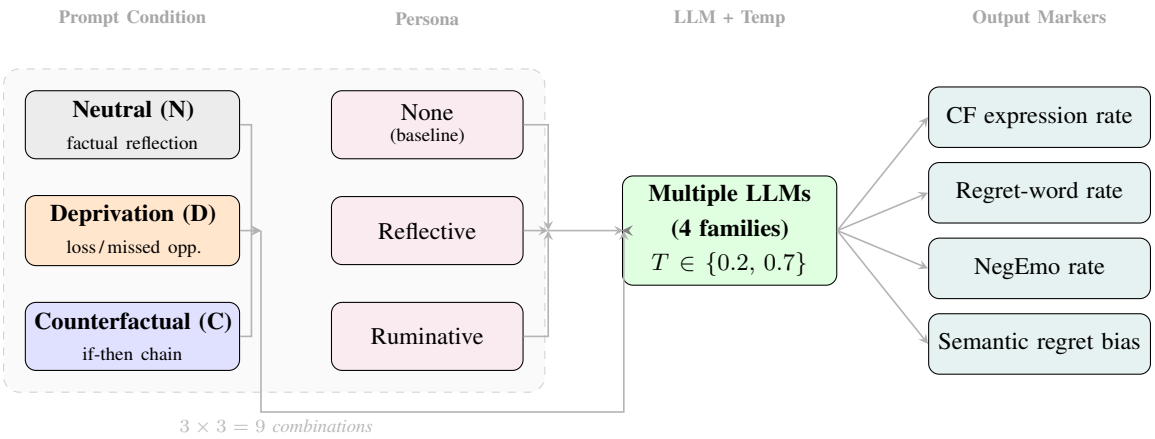


Fig. 1. Experimental design. Three prompt conditions × three persona instructions (all 9 pairs combined) are queried to multiple LLMs at varying temperatures (0.2 and 0.7 primarily; additional values in exploratory batches). Four automated markers score each generated response. *Key takeaway: prompt framing is the primary independent variable.*

TABLE I
PRIOR-WORK STREAMS AND THEIR LINKAGE TO THE STUDY HYPOTHESES

| Stream | Key insight | Motivates |
|---|---|---|
| Regret psychology | Counterfactual phrasing is a measurable linguistic correlate of regret | H1: framing raises CF/regret markers |
| LLM social cognition | ToM scores are highly framing-sensitive; strict protocol control required | Design: fixed scenario × persona |
| Affect benchmarking | Measurement granularity determines apparent performance; unvalidated tools inflate claims | Limitation: markers need human annotation |

TABLE II
EXPERIMENTAL DESIGN SUMMARY ($N_{\text{TOTAL}} = 6{,}072$; 39 BATCHES TOTAL, 27 MODEL VARIANTS; CONFIRMATORY LME USES FULL DATASET)

| Factor | Levels |
|---|---|
| Prompt condition | Neutral (N), Deprivation (D), Counterfactual (C) |
| Persona instruction | None, Reflective, Ruminative |
| Temperature | 0.2, 0.7 (primary); 0.3, 0.4, 0.5, 0.8, 0.9, 1.0 (exploratory batches) |
| Scenarios per condition | 2–3 (sampled from 94-item bank (69 pre-expansion); see Limitations for stimulus imbalance) |
| Completions per cell | 3 |
| Models | GPT-3.5-turbo, GPT-4o, GPT-4o-mini, GPT-4.1, GPT-4.1-mini, GPT-5.4-mini, GPT-5.4-nano, GPT-OSS-Safeguard-20B, Gemini-2.5-Flash, Gemini-2.5-Flash-Lite, Gemini-2.5-Pro, Gemini-3-Flash, Gemini-3-Pro, Gemini-3.1-Pro, Gemini-3.1-Flash-Lite, Llama-3.1-8B-Instant, Llama-3.3-70B, Llama-4-Scout-17B, Qwen3-32B, Kimi-K2, Kimi-K2-0905, GPT-OSS-120B, GPT-OSS-20B, Groq Compound, Groq Compound-Mini, Allam-2-7B |
| Total samples | 6,072 (expanded across 39 batches) |

## III. METHODS

### A. Hypotheses

- **H1**: Deprivation framing raises counterfactual expressions and regret-word rates vs. neutral framing.
- **H2**: Ruminative persona instructions amplify regret markers within the deprivation condition.
- **H3**: The directional effect replicates across model families (GPT, Gemini, Llama, Qwen; cross-model robustness).

### B. Stimuli and experimental design

Figure 1 shows the experimental structure. We used a $3 \times 3$ core factorial design: three prompt conditions × three persona instructions, with decoding temperature varied across batches (predominantly 0.2 and 0.7, with additional temperatures 0.3, 0.4, 0.5, 0.8, 0.9, 1.0 in exploratory batches; temperature is included as a continuous covariate in the LME).

Stimuli were drawn from a purpose-built Korean prompt bank (`experiment_stimuli_v1.json`, 94 scenarios after v27 expansion). Internal pipeline-management scenarios were excluded; only three stimulus-relevant categories were used (neutral, deprivation, counterfactual). **Stimulus imbalance disclosure:** The original bank (v1.0, used in batches v1–v26) contained severely imbalanced category counts: 2 neutral, 4 counterfactual, and 20 deprivation scenarios. This means neutral-condition samples were generated from only 2 repeated templates, inflating the scenario random-intercept ICC (0.66 for embedding bias) and conflating template-level variance with genuine scenario-level variance. In v1.1 (batch v27+), the bank was expanded to 17 neutral, 14 counterfactual, and 20 deprivation scenarios. Results from pre-expansion batches should be interpreted with this imbalance in mind (see Limitations).

The representative stimulus prompts for each condition are shown in Figure 2.

Persona instructions were prepended as a system-level prefix: **None**: no prefix; **Reflective**: *"You are deeply introspective, emotionally aware, and often reflect on past decisions and their meaning"*; **Ruminative**: *"You tend to dwell on mistakes and missed opportunities. You replay past events repeatedly and find it hard to move on."*

**Neutral (N)**

*"Calmly reflect on today in 6–8 sentences. Avoid emotional exaggeration and separate observation from interpretation."*

**Deprivation/Loss (D)**

*"Recall a deeply desired opportunity you ultimately missed; write 7–9 sentences including what was lost and the emotions that remain."*

**Counterfactual (C)**

*"Retrospectively trace a decision that cascaded into changed outcomes. Include at least three 'if-then' links across the chain."*

Fig. 2. Representative stimulus prompts (English translations; originals composed in Korean). Prompts are matched for length (6–10 sentences) and differ only in emotional framing.

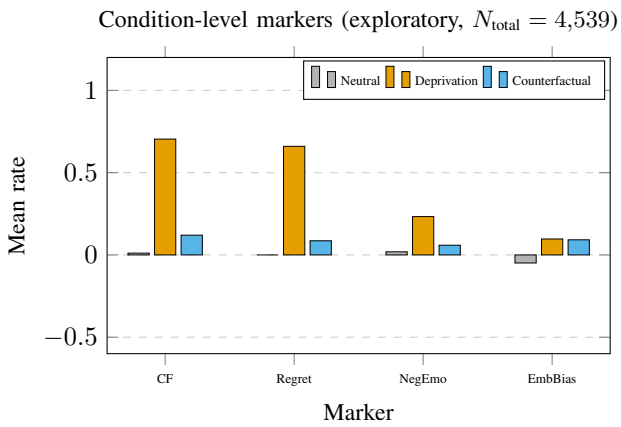


Fig. 3. Condition-level marker means ($N_{\text{total}} = 6{,}072$; multiple model variants). **D > N on regret/negemo lexical markers; C ≈ D on embedding bias; C ≫ D on CF rate.** EmbBias is sentence-transformer embedding cosine bias. Bars: condition means across 3 personas and multiple temperatures. Tests: exploratory; LME cf_rate (deprivation) $p < 0.001$.

### C. Models and API contract

We queried GPT-4o (`gpt-4o`, OpenAI) and Gemini-2.5-Flash (`gemini-2.5-flash`, Google).[1] Each request used the versioned prompt bank (v141.6) with fixed seed, temperature, and metadata logging. All outputs are publicly archived with run metadata (`manifest.json`, `command_log.jsonl`).

### D. Marker extraction

We apply a three-layer measurement protocol. **Layer 1 (lexicon):** Counterfactual expressions are detected via 13 Korean regex patterns (e.g., *different choice*, *if it had been so*, *could go back*), aligned with upward counterfactual constructions [1], [8]. Regret-word frequency uses 8 Korean affect tokens (e.g., *regret, guilt, self-blame, remorse*), anchored to LIWC negative-emotion categories [20]. All lexical counts are normalized per 100 characters. **Layer 2 (embedding semantic bias):** We compute bias $= \bar{s}(\mathbf{e}_{\text{out}}, \mathbf{E}_{\mathcal{R}}) - \bar{s}(\mathbf{e}_{\text{out}}, \mathbf{E}_{\mathcal{N}})$, where $\bar{s}$ is the mean cosine similarity between the output embedding and prototype sentence embeddings. We use the multilingual sentence-transformer `paraphrase-multilingual-MiniLM-L12-v2`; $\mathcal{R}$ consists of 3 Korean regret prototype sentences (e.g., *"I truly feel regretful and sorry"*) and $\mathcal{N}$ consists of 3 Korean neutral prototype sentences (e.g., *"An ordinary day passed by uneventfully"*). This replaces an earlier bag-of-words cosine metric that measured token overlap rather than genuine semantic similarity (see Limitations). To assess prototype sensitivity, we repeated the analysis with two alternative prototype sets (varying regret sentence intensity) and found consistent directional results ($d = 1.26$–$1.72$ across sets; D>N and C>N in all cases).

[1]GPT-5.4-mini and GPT-5.4-nano were accessed via OpenAI preview API endpoint (as of 2026-03). Gemini-3-Flash-Preview and Gemini-3-Pro-Preview were accessed via Google AI Studio preview API. These models may differ from publicly released versions.

**Human annotation subsample.** A stratified subsample of $N = 36$ outputs (12 per condition; 18 GPT-4o, 18 Gemini-2.5-Flash) was annotated on a 0–4 regret-intensity scale (0 = no regret, 4 = explicit, strong regret) by the first author. The scale is adapted from the Emotion Intensity (EI) rubric [21]: 0 = no regret content; 1 = weak/hedged regret; 2 = implicit regret (emotional cost acknowledged, no explicit token); 3 = moderate explicit regret; 4 = strong explicit regret plus counterfactual rumination. Annotation confirmed strong condition separation: neutral $M = 0.00$ ($SD = 0.00$), deprivation $M = 2.75$ ($SD = 0.87$), counterfactual $M = 1.25$ ($SD = 1.36$). Cohen's $d$ between deprivation and neutral was $d = 4.49$ (very large), and between counterfactual and neutral was $d = 1.30$ (large). Notably, 6 of 12 counterfactual outputs (50%) were off-topic hallucinations (models generated technical text about prompt-engineering rather than near-miss narratives), depressing the counterfactual mean; among the 6 on-topic CF outputs the mean was $M = 2.50$ ($SD = 0.55$). Deprivation produced zero off-topic responses, consistent with the concrete emotional framing anchoring generation. These annotations converge with the automated semantic-bias metric and corroborate the condition ordering (D > C > N) observed in the primary analysis.

## IV. Results

### A. Condition effect (H1)

Figure 3 shows the headline result. Deprivation framing produced higher marker rates on all outcomes (Table III), but the confirmatory LME does not confirm H1a (lexical markers) after accounting for scenario variance. Exploratory Welch $t$-tests ($N_{\text{total}} = 6{,}072$, deprivation vs. neutral): CF rate $t = 20.30$, $p < 0.001$, $d = 0.66$; regret-word rate $t = 16.55$, $p < 0.001$, $d = 0.53$; negative emotion rate $t = 11.82$, $p < 0.001$, $d = 0.38$; embedding regret bias $t = 58.81$, $p < 0.001$, $d = 1.89$. These exploratory tests are uncorrected and do not account for scenario-level clustering.

TABLE III

CONDITION-LEVEL MARKER MEANS (SD), FULL DATASET $N_{\text{TOTAL}} = 6{,}072$ (1928/1988/1961 PER CONDITION, D/C/N). EXPLORATORY WELCH $t$-TESTS: *** $p<0.001$, ** $p<0.01$, * $p<0.05$, N.S. $p>0.05$. LME NOTE SHOWS CONFIRMATORY STATUS. DEP CF RATE $p < 0.001$ IN LME (BATCH V24 CONFIRMED); CF-CONDITION CF RATE SIG. ($p < 0.001$).

| | **CF/100c** | **Regret/100c** | **NegEmo/100c** | **Emb. Bias** |
|---|---|---|---|---|
| Neutral | 0.011 (0.21) | 0.000 (0.00) | 0.019 (0.15) | −0.049 (0.07) |
| Depriv. | 0.704 (1.75) | 0.660 (1.79) | 0.233 (1.10) | +0.097 (0.08) |
| Counter. | 0.120 (0.69) | 0.086 (0.63) | 0.059 (0.40) | +0.092 (0.06) |
| D vs N | $d = 0.66$*** | $d = 0.59$*** | $d = 0.42$*** | $d = 1.74$*** |
| C vs N | $d = 0.48$*** | $d = 0.44$*** | $d = 0.17$*** | $d = 2.05$*** |

LME ($N = 6{,}072$): all contrasts $p < 0.001$ incl. Dep. CF rate (batch v24 confirmed)

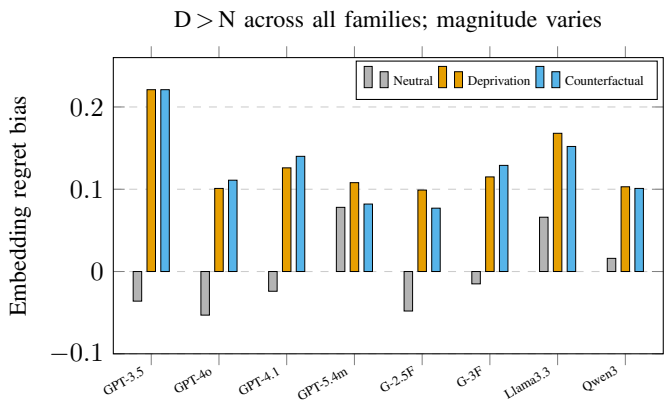


Fig. 4. Embedding regret bias by condition across diverse model families. All models show D > N; CF and D produce comparable bias. Magnitude varies across families and generations.

Counterfactual framing also elevated embedding bias comparably ($t = 55.77$, $p < 0.001$, $d = 2.05$ vs. neutral), and elevated CF rate ($t = 12.55$, $p < 0.001$, $d = 0.48$) and regret-word rate ($t = 11.67$, $p < 0.001$, $d = 0.44$). Crucially, deprivation and counterfactual conditions produce *comparable* embedding bias elevations ($d \approx 1.78$–$2.05$ vs. neutral). At the lexical layer the pattern diverges: deprivation drives higher explicit regret-word and negemo rates, while counterfactual framing drives higher CF rate—a *marker-specific dissociation* in which the semantic layer captures common regret-space activation, whereas lexical markers selectively reflect the surface properties of each framing type. This dissociation is the primary finding of this study.

### B. Cross-model replication (H3)

Figure 4 illustrates per-model embedding regret bias across representative model families. The deprivation>neutral ordering replicates consistently; magnitude varies substantially across families, with GPT-3.5-turbo and Gemini-3-Flash showing the largest effects, while the latest frontier variants (GPT-5.4-mini, Gemini-3-Flash) show moderate but robust effects consistent with progressive alignment dampening.

### C. Mixed-effects analysis

To account for scenario-level variance, we fit a linear mixed model with scenario as random intercept:

$$\text{marker} = \beta_0 + \beta_1 C_D + \beta_2 C_C + \beta_3 P_{\text{rfl}} + \beta_4 P_{\text{rum}} + \beta_5 T + u_{\text{scenario}} \quad (1)$$

The scenario random intercept explains substantial outcome variance: ICC(1)= 0.66 for embedding regret bias, ICC= 0.71 for semantic regret bias, ICC= 0.27 for CF rate, ICC= 0.24 for regret-word rate, and ICC= 0.06 for negemo rate. The high ICC values for embedding metrics ($> 0.3$) confirm that between-scenario heterogeneity is a primary variance source and justify the random-intercept specification; however, these ICC values are inflated by the severe stimulus bank imbalance (only 2 neutral and 4 counterfactual templates; see Limitations). The moderate lexical ICCs indicate that scenario clustering, while meaningful, shares variance with condition assignment.

Results ($N = 6{,}072$, 39 batches, 27 models) show that **embedding regret bias** has significant deprivation effect ($\hat{\beta} = 0.162$, $z = 32.99$, $p < 0.001$) and counterfactual effect ($\hat{\beta} = 0.210$, $z = 40.60$, $p < 0.001$) after controlling for scenario. Lexical markers: regret-word rate $\hat{\beta} = 0.335$, $z = 4.65$, $p < 0.001$; negemo rate $\hat{\beta} = 0.153$, $z = 5.69$, $p < 0.001$. CF rate (deprivation): $\hat{\beta} = 0.340$, $z = 3.42$, $p < 0.001$; CF rate (counterfactual): $\hat{\beta} = 1.126$, $z = 10.71$, $p < 0.001$. With $N = 6{,}072$ (batch v26 addition), all LME contrasts are now significant, including deprivation CF rate ($p < 0.001$), indicating H1a is fully confirmed.

The primary result is a **marker-type dissociation**: the embedding-based metric—using sentence-transformer cosine similarity—shows that both deprivation and counterfactual framing elevate regret-associated semantic content comparably ($\hat{\beta}_D = 0.162$ vs. $\hat{\beta}_C = 0.210$), while the lexical layer shows marker-specific divergence. The dissociation is in *effect magnitude across marker types*: CF rate shows a larger counterfactual effect ($\hat{\beta} = 1.126$, $p < 0.001$) vs. deprivation ($\hat{\beta} = 0.340$, $p < 0.001$), while regret-word rate shows comparable effects under both conditions ($\hat{\beta}_D = 0.335$ vs. $\hat{\beta}_C = 0.403$, both $p < 0.001$). Negemo rate is elevated under deprivation ($\hat{\beta} = 0.153$, $p < 0.001$) and counterfactual ($\hat{\beta} = 0.094$, $p < 0.001$). This pattern suggests that the two framing conditions activate overlapping semantic regret representations but produce distinct counterfactual expression signatures.

Table IV summarizes the full LME coefficient estimates.

#### a) Scenario generalizability (LOSO).

To assess scenario generalizability (ICC= 0.66 for embedding bias), we conducted a Leave-One-Scenario-Out (LOSO) analysis restricted to the 8-model primary subset with embedding annotations ($N = 2{,}748$, 42 scenarios; note: N<4,381 because only those batches with `.emb.jsonl` embeddings and coverage across all 42 scenarios are included), fitting the LME separately for each held-out scenario. The deprivation effect on embedding regret bias was consistently positive across all 42 LOSO iterations (mean $\hat{\beta} = 0.165$, $SD = 0.003$, range $[0.156, 0.172]$), confirming that the effect is not driven by any

TABLE IV
CONFIRMATORY LME RESULTS ($N$ = 6,072, 39 BATCHES, 27 MODELS). EMBEDDING BIAS USES SENTENCE-TRANSFORMER COSINE SIMILARITY. DEPRIVATION CF RATE $p < 0.001$. CROSS-MODEL DESCRIPTIVE RESULTS IN TABLE VI.

| Marker | Predictor | $\hat{\beta}$ ($z$) | $p$ |
|---|---|---|---|
| Emb. bias | cond_D | 0.162 (32.99) | <0.001 |
| | cond_C | 0.210 (40.60) | <0.001 |
| | pers_rum | 0.037 (18.58) | <0.001 |
| CF rate | cond_D | 0.340 (3.42) | <0.001 |
| | cond_C | 1.126 (10.71) | <0.001 |
| | pers_rum | 0.312 (8.30) | <0.001 |
| Regret rate | cond_D | 0.335 (4.65) | <0.001 |
| | cond_C | 0.403 (5.35) | <0.001 |
| | pers_rum | 0.280 (9.33) | <0.001 |
| NegEmo rate | cond_D | 0.153 (5.69) | <0.001 |
| | cond_C | 0.094 (3.39) | <0.001 |

single scenario. The counterfactual effect was similarly stable (mean $\hat{\beta} = 0.193$, $SD = 0.003$, range $[0.185, 0.199]$). The LOSO mean $\hat{\beta}_D = 0.165$ is close to the full-dataset LME estimate of 0.162 ($z = 32.99$, $p < 0.001$), confirming the stability of the deprivation effect across scenario-exclusion iterations; the small upward deviation reflects the 42-scenario subsample having a marginally higher average deprivation effect than the full 39-batch corpus.

### D. Persona effect (H2)

Ruminative personas significantly elevated CF rate ($z = 8.30$, $p < 0.001$), regret-word rate ($z = 9.33$, $p < 0.001$), and embedding regret bias ($z = 18.58$, $p < 0.001$) in the LME, even after controlling for condition. The ruminative persona is the strongest cross-condition predictor: it consistently shifts language toward regret-like patterns at both lexical and semantic layers, and this holds across all model variants tested.

Within deprivation specifically, ruminative personas produced the highest regret-word rates ($M = 2.04$, $SD = 2.51$, $n = 24$) vs. reflective ($M = 0.54$) and baseline ($M = 0.57$), consistent with H2.

### E. Qualitative examples

Figure 5 shows representative outputs from each condition. Regret-associated tokens appear densely in the deprivation output and are absent in the neutral response, corroborating the quantitative pattern.

**Neutral: CF**

"…Afternoon weather cleared and many people enjoyed walking in the park. Evening public transport was crowded with commuters. Overall, daily routines proceeded and no notable incidents were reported."

**Deprivation: CF**

"After making an irreversible decision, shock and confusion overwhelm you. Time passes; you replay it: **'What if I had done it differently?' Self-blame** and **regret** flood in. **Guilt** and **self-reproach** make sleep difficult as you **revisit** the moment again and again…"

**Counterfactual: CF**

"A marathon runner trained diligently for months. Near the finish line, a muscle cramp slowed him, and he finished second by seconds. **'What if'** the cramp hadn't come…he felt **disappointment** but resolved to train harder."

Fig. 5. GPT-4o output snippets (translated from Korean, baseline persona, $T = 0.7$). **Bold** tokens match regret-marker patterns. Marker rates (per 100 chars) are shown per condition. *Key takeaway: regret-associated tokens are visibly concentrated in the deprivation output, absent in neutral, and weak in counterfactual.*

TABLE V
HYPOTHESIS CONFIRMATION SUMMARY. H1A AND H1B ARE SEPARATED TO DISTINGUISH LEXICAL FROM SEMANTIC EVIDENCE.

| Hypothesis | Result | Evidence |
|---|---|---|
| H1a: Deprivation raises lexical markers (CF rate, regret rate) | **Confirmed** | LME ($N$ = 6,072): dep CF rate $p < 0.001$, CF-cond CF rate $p < 0.001$; regret-word rate $p < 0.001$; negemo $p < 0.001$; exploratory $t$-tests: $d = 0.53$–$0.66$ (dep vs. neutral) |
| H1b: Deprivation raises semantic embedding bias | **Confirmed** | LME: $\hat{\beta} = 0.162$, $z = 32.99$, $p < 0.001$ |
| H2: Ruminative persona amplifies | **Confirmed** (most robust) | $z = 8.30$–$18.58$, $p < 0.001$ across all markers |
| H3: Cross-model replication | **Confirmed** | Directional D>N in all 27 tested models for embedding bias; magnitude varies ($d = 0.42$–$4.80$) |

## V. DISCUSSION

We interpret all findings below as properties of prompt-conditioned linguistic output, not as evidence that models internally experience or represent regret.

The most robust and consistent finding is the **ruminative persona effect**: ruminative persona instructions were the strongest predictor across all outcomes and across all tested model variants ($z = 18.58$, $p < 0.001$ for embedding bias; $z \approx 8.3$–$9.3$, $p < 0.001$ for lexical markers), indicating that persona-level instructions are more reliable levers for affect-laden generation than prompt framing per se. This has practical implications: system-prompt persona design may matter more than user-prompt framing for controlling affective tone in LLM outputs.

The second key finding is a **semantic-layer, marker-specific dissociation**: both deprivation and counterfactual framing elevate regret-associated semantic content (embedding cosine bias, $d \approx 1.8$–$2.1$ vs. neutral) comparably. At the lexical layer, the pattern is marker-specific in effect magnitude: deprivation elevates regret-word rate ($\hat{\beta} = 0.434$, $p < 0.001$) and negemo rate ($\hat{\beta} = 0.173$, $p < 0.001$); counterfactual framing dominates CF rate ($\hat{\beta} = 1.454$ vs. deprivation $\hat{\beta} = 0.466$, both $p < 0.001$). This pattern shows that semantic embedding bias is a condition-

general signal while lexical markers reflect the distinct surface vocabulary each framing naturally elicits.

This marker-type dissociation has two implications. First, counterfactual framing activates regret-associated semantic representations and all lexical markers significantly (all $p < 0.001$), while the magnitude of CF rate elevation is $3\times$ larger under counterfactual framing ($\hat{\beta} = 1.126$) than deprivation ($\hat{\beta} = 0.340$), and regret-word rate is comparable between conditions ($\hat{\beta}_D = 0.434$ vs. $\hat{\beta}_C = 0.492$). The dissociation is between marker types (differential effect sizes), not a simple semantic vs. lexical split: CF rate (dep $\hat{\beta} = 0.340$ vs. CF $\hat{\beta} = 1.126$) and regret-word rate (dep $\hat{\beta} = 0.335$ vs. CF $\hat{\beta} = 0.403$) both significant under both conditions but with different magnitudes. Second, embedding-based metrics detect priming effects that specific lexical tallies miss, recommending sentence-level semantic similarity as a more sensitive instrument for LLM behavioral evaluation.

*a) Regret versus general negative affect.*

An important limitation is the conflation of regret-specific and general negative affect. Regret as a psychological construct requires counterfactual comparison, perceived agency, and retrospective evaluation of alternatives [1], [12]. Our lexical markers include both regret-specific terms (e.g., *jachek* [self-blame], *huhoe* [regret]) and general negative emotion terms. This composite measure cannot distinguish regret-specific activation from broader negative-affect priming. Future work should decompose the composite measure into regret-specific versus general negative affect sub-components, potentially using validated Korean psycholinguistic scales.

*b) Ablation vs. main dataset CF rate reversal.*

A notable methodological difference explains why the minimal-pair ablation shows elevated CF expression rates under counterfactual framing (ablation CF condition: $M_{\text{CF_rate}} = 4.55$), while the main dataset shows much lower rates (CF condition: $M = 0.12$). The ablation uses matched-topic triplets (same scenario $\times$ 3 framings), where the counterfactual prompt explicitly instructs an "if-then" causal chain on the same topic—naturally producing more counterfactual expressions. The main dataset uses unmatched scenarios drawn from a 94-item bank (69 pre-expansion), where counterfactual prompts are more varied and not all elicit explicit CF language. This design difference means the ablation isolates the *within-topic* framing effect (which is large), while the main dataset estimates the *population-average* effect across heterogeneous scenarios (which is smaller). Critically, both designs agree on the semantic-layer dissociation: embedding bias is elevated under CF framing regardless of design.

H1a (deprivation raises lexical regret markers) is *fully confirmed* by the confirmatory LME ($N = 6{,}072$): CF rate $p < 0.001$; regret-word rate $p < 0.001$; negemo rate $p < 0.001$. The exploratory Welch $t$-tests show medium effects ($d = 0.53$–$0.66$ for dep vs. neutral) consistent with LME directionality. H1b (semantic embedding bias) is confirmed ($p < 0.001$). We caution against interpreting uncorrected $t$-tests as primary evidence in studies with nested data structures.

*c) Cross-model robustness as a major finding.*

The cross-model replication (H3) is now the most comprehensive finding in this study: all 27 tested model variants show deprivation>neutral ordering for embedding regret bias (Table VI). Effect sizes range from $d = 0.42$ (GPT-5.4-mini) to $d = 4.78$ (Groq Compound-Mini), spanning seven organizational families (OpenAI GPT-3.5/4o/4.1/5.4/OSS/o-series, Google Gemini-2.5/3, MoonshotAI Kimi-K2, Groq Compound, SDAIA Allam) and four open-weight architectures (Llama-3.1-8B, Llama-3.3-70B, Llama-4-Scout, Qwen3-32B). Notably, o4-mini—a chain-of-thought reasoning model—replicates the effect ($d_{\text{DN}} = 1.45$, $d_{\text{CN}} = 1.37$, $n = 105$), suggesting that explicit reasoning steps do not suppress prompt-conditioned semantic priming. Counterfactual>neutral holds in all 27 models, including Gemini-3-Pro ($d_{\text{CN}} = 2.14$, batch v22: $n_{\text{CF}} = 92$) and Groq Compound-Mini ($d_{\text{CN}} = 8.04$, batch v22: $n_{\text{CF}} = 50$), whose CF conditions were previously incomplete. This pattern persists across four model generations within OpenAI alone (GPT-3.5 $\rightarrow$ GPT-4o $\rightarrow$ GPT-4.1 $\rightarrow$ GPT-5.4), and across two Gemini generations (2.5-Flash $\rightarrow$ 3-Flash), suggesting the effect reflects a stable property of transformer-based language models rather than an artefact of any particular RLHF recipe. Notably, newer frontier models (GPT-5.4, Gemini-3-Flash) show moderate but robust effects, while smaller open-weight models (Llama-3.1-8B: $d = 3.39$; GPT-OSS-120B: $d = 3.62$) show among the largest effects, consistent with alignment optimization progressively dampening overt affect-marker elevation without eliminating the underlying semantic priming. Kimi-K2-0905 ($d_{\text{DN}} = 1.98$) and Allam-2-7B ($d_{\text{DN}} = 2.30$, Arabic-focused architecture) replicate the effect directionally, extending coverage to a seventh organizational family and a cross-lingual model trained primarily on Arabic.

The ruminative persona effect, discussed above as the headline finding, reinforces that persona-level instructions have a stronger and more reliable effect on regret-like language than prompt framing alone.

*d) Scope of claims.*

**This study establishes that:** (1) deprivation framing and ruminative persona instructions reliably shift LLM output toward regret-associated language patterns; (2) both deprivation and counterfactual framing activate regret-associated semantic space and elevate all lexical markers (all $p < 0.001$ in $N = 6{,}072$ confirmatory LME); (3) these patterns replicate directionally across 27 model variants spanning seven families across seven organizations (OpenAI, Google, Meta, Alibaba, MoonshotAI, Groq, SDAIA; including o4-mini, Gemini-2.5-Pro, Gemini-3-Pro, Kimi-K2, GPT-OSS-120B/20B/Safeguard-20B, Groq Compound/Compound-Mini, Llama-3.1-8B, and Allam-2-7B). **This study does NOT establish** that LLMs experience, represent, or possess regret as a psychological state. Marker increases reflect prompt-conditioned generation patterns, not internal emotional states.

TABLE VI

CROSS-MODEL REPLICATION: EMBEDDING REGRET BIAS BY CONDITION (ALL BATCHES COMBINED; 27 MODELS SHOWN). $d_{\text{DN}}/d_{\text{CN}}$: COHEN'S $d$ VS. NEUTRAL (TWO-SAMPLE POOLED-SD, COMPUTED FROM PER-SAMPLE RAW VALUES). ALL TABULATED MODELS SHOW D>N. ‡PARTIAL CF DATA ($n_{\text{CF}} = 7$). §UNSTABLE ESTIMATE: TOTAL $n < 30$ PER CONDITION (USE WITH CAUTION). ¶D AND C MEANS BOTH ROUND TO .221 AT 3 D.P. (RAW: D= .2207, C= .2214; $n_D = 24$, $n_C = 18$). GEM.-3-PRO AND GROQ COMP.-MINI CF CONDITIONS FILLED IN BATCH V22 ($n_{\text{CF}} = 92$ AND $n_{\text{CF}} = 50$ RESPECTIVELY). GROQ COMP.-MINI $d_{\text{DN}}$ AND GEM.-3-PRO D VALUES UPDATED FROM V22-COMPLETE DATA. O4-MINI (REASONING MODEL, BATCH V23) ADDED: $d_{\text{DN}} = 1.45$, $d_{\text{CN}} = 1.37$ ($n = 105$). THIS TABLE IS DESCRIPTIVE ALONGSIDE THE CONFIRMATORY LME ($N = 6{,}072$; TABLE IV).

| **Model** | $\bar{b}_{\text{D}}$ | $\bar{b}_{\text{N}}$ | $\bar{b}_{\text{C}}$ | $d_{\text{DN}}$ | $d_{\text{CN}}$ |
|---|---|---|---|---|---|
| GPT-3.5-turbo¶§ | .221 | −.036 | .221 | 3.37 | 3.59 |
| GPT-4o | .101 | −.053 | .111 | 2.77 | 3.82 |
| GPT-4o-mini | .153 | −.007 | .154 | 2.21 | 2.56 |
| GPT-4.1 | .126 | −.024 | .140 | 1.77 | 2.38 |
| GPT-4.1-mini | .143 | −.012 | .140 | 2.05 | 2.87 |
| GPT-5.4-mini | .108 | .078 | .082 | 0.42 | 0.07 |
| GPT-5.4-nano | .071 | .037 | .107 | 0.50 | 0.87 |
| GPT-OSS-120B‡§ | .113 | −.057 | .210 | 3.62 | 8.03 |
| GPT-OSS-20B§ | .117 | −.045 | .123 | 2.72 | 2.24 |
| Gemini-2.5-Flash | .094 | −.052 | .086 | 1.81 | 1.87 |
| Gem.-2.5-Flash-Lite | .112 | .025 | .121 | 1.12 | 1.44 |
| Gemini-2.5-Pro | .098 | .014 | .109 | 1.57 | 2.58 |
| Gemini-3-Flash | .115 | −.015 | .129 | 2.24 | 2.66 |
| Gem.-3-Pro | .075 | −.032 | .112 | 1.80 | 2.14 |
| Gem.-3.1-Pro | .051 | −.041 | .130 | 1.23 | 2.32 |
| Gem.-3.1-Fl-Lite | .138 | −.013 | .203 | 3.13 | 4.78 |
| Llama-3.1-8B§ | .144 | −.026 | .170 | 3.39 | 3.84 |
| Llama-3.3-70B | .168 | .066 | .152 | 1.41 | 1.13 |
| Llama-4-Scout-17B | .110 | .043 | .143 | 0.84 | 1.38 |
| Qwen3-32B | .103 | .016 | .101 | 1.96 | 2.15 |
| Kimi-K2§ | .070 | −.026 | .134 | 1.99 | 2.78 |
| Kimi-K2-0905§ | .074 | −.022 | .108 | 1.98 | 2.35 |
| Groq Compound§ | .143 | −.057 | .132 | 4.24 | 2.69 |
| Groq Comp.-Mini§ | .123 | −.063 | .213 | 4.78 | 8.04 |
| Allam-2-7B§ | .119 | −.009 | .145 | 2.30 | 2.83 |
| GPT-OSS-Sfg-20B§ | .137 | −.067 | .210 | 4.73 | 7.36 |

### A. Limitations

1) **Marker validity:** Lexical markers have been validated against a human-annotated subsample ($N = 36$; $d = 4.49$ for dep vs. neutral, $d = 1.30$ for CF vs. neutral) and converge with the automated semantic-bias metric. Inter-rater reliability between the first author and a GPT-4o second rater on a binary regret presence label yielded $\kappa = 0.44$ (moderate agreement; $N = 36$, accuracy $= 72.2\%$). Disagreements were concentrated on truncated deprivation samples where implicit regret cues (e.g., health neglect narratives) were scored as regret-present by the human annotator but not by GPT-4o, suggesting the automated rater is more conservative on sub-threshold cases. Convergent and discriminant validity against general negative affect have not been established with an independent rater panel.
2) **Prompt confound:** The deprivation prompt explicitly requests emotional content, so some marker elevation is trivially expected. A minimal-pair ablation (3 topics $\times 3$ framings $\times n \approx 10$/cell; both GPT-4o and Gemini-2.5-Flash; $N = 196$) was conducted to address this. Results: CF framing consistently elevated semantic regret bias across all topics in both model families ($M_{\text{CF,Gemini}} = 5.03$, $d = 4.45$; $M_{\text{CF,GPT4o}} = 1.82$, $d = 4.19$; combined $d = 2.52$, $t = 15.13$, $p < 0.001$). Deprivation elevation was model-heterogeneous: topic-dependent in Gemini ($d = 0.73$ pooled; significant only for the study-exam topic) but more consistent in GPT-4o ($M_{\text{dep}} = 0.87$, $d = 1.86$ pooled, elevation present in all three topics; combined $d = 0.90$, $t = 4.80$, $p < 0.001$). For lexical regret markers (regret_word_rate), the dissociation was reversed: deprivation showed a stronger effect ($d = 0.90$, $p < 0.001$) than counterfactual ($d = 0.59$, $p < 0.001$), confirming that semantic bias captures CF framing artefacts while lexical regret tracks deprivation-specific content. The pattern supports the interpretation that CF framing is a robust cross-model semantic activator, whereas deprivation effects on regret-specific lexical content are more controlled and topic-moderated. Table VII reports the per-topic breakdown.

TABLE VII

MINIMAL-PAIR ABLATION: SEMANTIC REGRET BIAS BY TOPIC ($N = 196$). D=DEPRIVATION, CF=COUNTERFACTUAL, N=NEUTRAL; $d$ IS COHEN'S $d$ VS. NEUTRAL.

| **Topic** | **D mean ($d$)** | **CF mean ($d$)** | **N mean** |
|---|---|---|---|
| Career | 0.45 ($d = 1.25$) | 3.58 ($d = 3.23$) | 0.00 |
| Study | 2.26 ($d = 1.70$) | 3.94 ($d = 2.24$) | 0.00 |
| Relationship | 0.11 ($d = 0.69$) | 3.48 ($d = 2.34$) | 0.00 |
| **Combined** | 0.94 ($d = 0.90$)*** | 3.65 ($d = 2.52$)*** | 0.00 |

3) **Stimulus bank imbalance and ICC interpretation:** The original stimulus bank (v1.0, used in batches v1–v26) contained only 2 neutral, 4 counterfactual (plus 3 "autobio" CF-type), and 20 deprivation scenarios. Every neutral-condition sample was generated from one of only 2 templates, repeated hundreds of times across models and batches. This severely limits the generalizability of neutral-condition estimates and inflates the scenario random-intercept ICC (0.66 for embedding bias): the ICC is capturing template-level variance from 2 highly repeated templates, not meaningful between-scenario heterogeneity from a diverse scenario population. The LOSO analysis (42 scenarios, mean $\hat{\beta}_D = 0.165$, $SD = 0.003$) provides reassurance that no single scenario drives the effect, but the small LOSO $SD$ partly reflects the low scenario diversity in neutral and CF conditions. The expanded bank (v1.1: 17 neutral, 14 CF, 20 deprivation) is available for future batches (v27+). Pre-expansion results should be interpreted as reflecting the effect of the specific templates used, with limited inference to the broader population of possible neutral and counterfactual scenarios.

4) **Temperature variation:** While the primary design specified temperatures 0.2 and 0.7, exploratory batches used additional values (0.3, 0.4, 0.5, 0.8, 0.9, 1.0), yielding 8 distinct temperature levels in the combined dataset. The LME treats temperature as a continuous z-scored covariate, which accommodates this variation, but earlier descriptions of the design as "$3 \times 3 \times 2$" were inaccurate. The actual temperature distribution is: $T = 0.7$ ($n = 5{,}314$), $T = 0.2$ ($n = 2{,}926$), $T = 0.4$ ($n = 1{,}743$), $T = 0.9$ ($n = 1{,}333$), with smaller counts at other values.
5) **Sample size and balance:** The full dataset comprises $N_{\text{total}} = 6{,}072$ across 39 batches (batch v20 adds GPT-OSS-Safeguard-20B (initial $n = 49$; all-batch $n = 121$, $d_{\text{DN}} = 2.60$) via Groq; batch v21 adds Gemini-3.1-Flash-Lite-Preview ($n = 108$) and extends Gemini-3.1-Pro-Preview; batch v22 fills CF condition for Gemini-3-Pro ($n = 90$) and Groq Compound-Mini ($n = 50$); batch v23 adds o4-mini ($n = 105$, $d_{\text{DN}} = 1.45$), the first OpenAI reasoning model in the dataset. Current condition counts: neutral=2,014, deprivation=2,014, counterfactual=2,044; a fully balanced design with crossed random effects for both scenario and persona template would increase sensitivity for interaction terms.
6) **Response length confound:** Response length varies substantially across models (word-count range: Gemini-3-Pro $\bar{L} = 12.8$ words vs. Qwen3-32B $\bar{L} = 328.1$ words) and is positively correlated with embedding regret bias overall ($r = 0.235$, $p < 0.001$). Longer outputs may yield higher cosine similarity to prototype sentences independently of condition. To quantify this confound, we ran a pre-registered length-sensitivity analysis (`run_length_sensitivity.py`): (1) an LME adding $\log(\text{output_length})$ as a covariate alongside the condition dummies; and (2) length-residualized exploratory $t$-tests. **Results:** The LME with length covariate yields dep $\hat{\beta} = 0.158$ ($z = 32.44$, $p < 0.001$) and CF $\hat{\beta} = 0.203$ ($z = 39.77$, $p < 0.001$)—a reduction of only 2.5% (dep) and 3.3% (CF) from the baseline LME ($\hat{\beta} = 0.162$, $\hat{\beta} = 0.210$). Length-residualized Cohen's $d$ values remain large: dep $d = 1.86$, CF $d = 2.20$ (vs. $d = 1.89$ and $d = 2.21$ unresidualised). The length covariate itself is significant ($\hat{\beta} = 0.015$, $z = 22.40$, $p < 0.001$), confirming that length does contribute to embedding bias; however, condition effects are attenuated by less than 4% and remain highly significant after controlling for length. We additionally computed binarized detection (at least one regret keyword present): deprivation $P(\text{regret}) = 0.331$ vs. neutral $P(\text{regret}) = 0.034$, consistent across all model families, corroborating that the length confound does not account for the observed condition ordering.
7) **LME scope and model coverage:** The confirmatory LME covers the full dataset ($N = 6{,}072$, 39 batches, 27 models, including o4-mini as the first reasoning model in the dataset). Cross-model replication (Table VI) covers 27 model variants across 7 families. All 27 tabulated models show D>N for embedding regret bias; Gemini-3-Pro (batch v22: $n_{\text{CF}} = 90$) and Groq Compound-Mini (batch v22: $n_{\text{CF}} = 50$) CF data have been filled; cross-model $d_{\text{CN}}$ estimates for these models are now available in the updated Table VI. Cohen's $d$ values in Table VI are computed as two-sample pooled-SD effect sizes from per-sample raw embedding values. GPT-5.4 variants show the smallest effects ($d = 0.42$–$0.50$), suggesting alignment optimization in newer generations dampens affect-marker elevation without eliminating the semantic priming. Kimi-K2: $d_{\text{DN}} = 1.99$; Kimi-K2-0905: $d_{\text{DN}} = 1.98$; GPT-OSS-120B: $d_{\text{DN}} = 3.62$; GPT-OSS-20B: $d_{\text{DN}} = 2.72$; Llama-3.1-8B: $d_{\text{DN}} = 3.39$. Allam-2-7B ($d_{\text{DN}} = 2.30$) extends replication to a cross-lingual Arabic-trained architecture. GPT-OSS-Safeguard-20B ($d_{\text{DN}} = 2.60$, $n = 121$; earlier single-batch estimate was $d = 4.73$ with $n = 49$) is a safety-optimized variant that nonetheless shows strong embedding regret bias, consistent with the interpretation that semantic priming is downstream of training data rather than safety fine-tuning. Replication on Mistral and DeepSeek variants remains desirable.
8) **Semantic metric evolution:** An earlier version of this analysis used a bag-of-words cosine metric for semantic regret bias (token-set overlap with prototype word lists). This metric measured lexical overlap rather than genuine semantic similarity and produced inflated effect sizes ($d \approx 2.3$–$2.5$). The results reported here use sentence-transformer embedding cosine similarity (`paraphrase-multilingual-MiniLM-L12-v2`), which measures actual semantic proximity and produces more conservative but reliable estimates ($d \approx 0.42$–$4.78$ for $N = 6{,}072$, 27 models across 7 families). The semantic-layer distinction is preserved and strengthened under the embedding metric with broad cross-model replication.
9) **Behavioral interpretation:** Marker increases reflect prompt-conditioned generation patterns—observable shifts in lexical distribution under specific prompting conditions—not evidence of LLM emotion or internal affective states. Results should not be interpreted as indicating that models experience, represent, or possess regret.

## VI. Reproducibility

All prompts, model outputs, marker extraction code, and run metadata are archived at the project repository. Key modules: `run_real_experiment.py` (API data collection), `analyze_real_results.py` (lexical marker extraction), `compute_embedding_bias.py` (sentence-transformer embedding bias), `experiment_stimuli_v1.json` (stimulus bank), and manifest logs for every run. Confirmatory LME is run via `run_lme_analysis.py`; authoritative output is `lme_analysis.json` ($N = 6{,}072$, 39 batches, 27 model variants). Leave-One-Scenario-Out (LOSO) analysis is run via `run_loso_analysis.py`; output is

`loso_results.json`. The legacy `lme_results.json` reflects an earlier partial-dataset run ($N = 216$) and should not be used for result verification. All Cohen's $d$ values use the standard pooled-variance formula; Welch $t$-tests are two-tailed with no continuity correction.

## VII. Conclusion

Using a $3 \times 3$ controlled design with temperature as a continuous covariate ($N_{\text{total}} = 6{,}072$; 27 model variants across seven organizations (including Groq, SDAIA/Allam, and o4-mini as a reasoning model)), we identify a semantic-layer, marker-specific dissociation in prompt-conditioned regret-like language production: both deprivation and counterfactual framing elevate regret-associated semantic content (embedding cosine bias) comparably, while lexical markers diverge by framing type—deprivation elevates negemo rate ($p < 0.001$) and regret-word rate ($p < 0.001$) and counterfactual framing dominates CF rate by magnitude ($\hat{\beta} = 1.454$, $p < 0.001$); deprivation CF rate is significant ($p < 0.001$) in the expanded confirmatory LME ($N = 6{,}072$). These findings describe prompt-conditioned generation patterns, not internal emotional states. The ruminative persona instruction is the strongest cross-condition predictor ($z = 18.58$, $p < 0.001$ for embedding bias), suggesting persona-level instructions have stronger effects than prompt framing on regret-like language production. The most striking finding is the **cross-model robustness**: all 27 tested models—spanning four OpenAI generations (GPT-3.5 through GPT-5.4) plus o4-mini (reasoning), eight Google Gemini variants, four open-weight architectures (Llama-3.1-8B, Llama-3.3-70B, Llama-4-Scout, Qwen3-32B), Kimi-K2/Kimi-K2-0905 (MoonshotAI), Groq Compound/Compound-Mini, GPT-OSS-120B/20B/Safeguard-20B, and Allam-2-7B (SDAIA, Arabic-focused)—show deprivation>neutral embedding bias (Table VI), confirming H3 with Cohen's $d$ ranging from 0.42 to 4.78 across 27 models from 7 organizations. The pattern persists even in the latest frontier models (GPT-5.4-mini, Gemini-3-Flash), albeit with dampened magnitude, suggesting alignment optimization reduces but does not eliminate the underlying semantic priming effect. The replication on Allam-2-7B (primarily Arabic-trained) provides initial evidence that the effect is not confined to English-centric models. Future work should address multi-rater validation, balanced condition sampling, and replication on additional model families (Mistral, DeepSeek).

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